Fatima Jinnah Women University

Submitted to:-

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Submitted by:-

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Reg no:-

2023-BSE-021

Section:-

5 A

Cloud Computing

Lab Exam

https://github.com/Hafsa-Khalid021/Lab-Exam

Q1 – AWS IAM Setup Using AWS CLI and Console Verification

```
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $ aws iam create-group --group-name Software
Engineering
{
    "Group": {
        "Path": "/",
        "GroupName": "SoftwareEngineering",
        "GroupId": "AGPA3ED5IU4DHJRECLPKS",
        "Arn": "arn:aws:iam::764766889734:group/SoftwareEngineering",
        "CreateDate": "2026-01-19T07:43:25+00:00"
    }
}
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $
```

```
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $ aws iam get-group --group-name SoftwareEng
ineering
{
    "Users": [],
    "Group": {
        "Path": "/",
        "GroupName": "SoftwareEngineering",
        "GroupId": "AGPA3ED5IU4DHJRECLPKS",
        "Arn": "arn:aws:iam::764766889734:group/SoftwareEngineering",
        "CreateDate": "2026-01-19T07:43:25+00:00"
    }
}
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $
```

```
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $ aws iam create-user --user-name Hafsa
{
    "User": {
        "Path": "/",
        "UserName": "Hafsa",
        "UserId": "AIDA3ED5IU4DNWZ3X25HI",
        "Arn": "arn:aws:iam::764766889734:user/Hafsa",
        "CreateDate": "2026-01-19T07:45:00+00:00"
    }
}
```

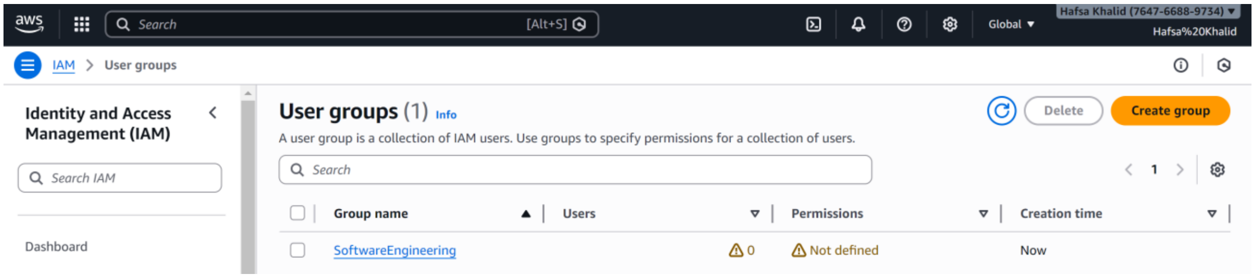
```
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $ aws iam add-user-to-group \
  --user-name Hafsa \
  --group-name SoftwareEngineering
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $
```

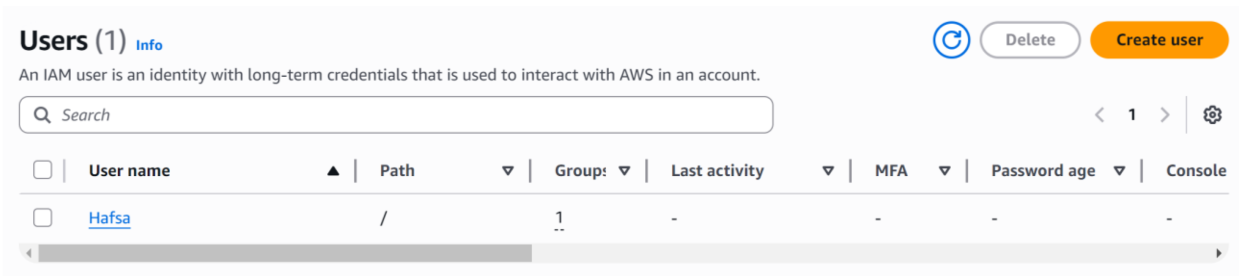
```
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $ aws iam get-group --group-name SoftwareEng
ineering
{
    "Users": [
        {
            "Path": "/",
            "UserName": "Hafsa",
            "UserId": "AIDA3ED5IU4DNWZ3X25HI",
            "Arn": "arn:aws:iam::764766889734:user/Hafsa",
            "CreateDate": "2026-01-19T07:45:00+00:00"
        }
    ],
    "Group": {
        "Path": "/",
        "GroupName": "SoftwareEngineering",
        "GroupId": "AGPA3ED5IU4DHJRECLPKS",
        "Arn": "arn:aws:iam::764766889734:group/SoftwareEngineering",
        "CreateDate": "2026-01-19T07:43:25+00:00"
    }
}
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $
```

```
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $ aws iam list-policies --scope AWS --query
"Policies[?PolicyName=='AdministratorAccess']"
[
    {
        "PolicyName": "AdministratorAccess",
        "PolicyId": "ANPAIWMBCKSKIEE64ZLYK",
        "Arn": "arn:aws:iam::aws:policy/AdministratorAccess",
        "Path": "/",
        "DefaultVersionId": "v1",
        "AttachmentCount": 0,
        "PermissionsBoundaryUsageCount": 0,
        "IsAttachable": true,
        "CreateDate": "2015-02-06T18:39:46+00:00",
        "UpdateDate": "2015-02-06T18:39:46+00:00"
    }
]
```

```
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $ aws iam attach-group-policy \
  --group-name SoftwareEngineering \
  --policy-arn arn:aws:iam::aws:policy/AdministratorAccess
```

```
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $ aws iam list-attached-group-policies \
  --group-name SoftwareEngineering
{
    "AttachedPolicies": [
        {
            "PolicyName": "AdministratorAccess",
            "PolicyArn": "arn:aws:iam::aws:policy/AdministratorAccess"
        }
    ]
}
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $
```





Q2 – Terraform Lab: Simple AWS Environment with Nginx over HTTPS

```
@Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $ terraform init
Initializing the backend...
Initializing provider plugins...
- Finding latest version of hashicorp/http...
- Reusing previous version of hashicorp/aws from the dependency lock file
- Installing hashicorp/http v3.5.0...
- Installed hashicorp/http v3.5.0 (signed by HashiCorp)
- Using previously-installed hashicorp/aws v6.28.0
Terraform has made some changes to the provider dependency selections recorded
in the .terraform.lock.hcl file. Review those changes and commit them to your
version control system if they represent changes you intended to make.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
```

```
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $ terraform plan
      + ipv6_cidr_block                      = (known after apply)
      + ipv6_cidr_block_network_border_group = (known after apply)
      + main_route_table_id                  = (known after apply)
      + owner_id                             = (known after apply)
      + region                               = "me-central-1"
      + tags                                 = {
          + "Name" = "dev-vpc"
        }
      + tags_all                             = {
          + "Name" = "dev-vpc"
        }
    }

Plan: 7 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + ec2_public_ip = (known after apply)

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Note: You didn't use the -out option to save this plan, so Terraform can't guarantee to
take exactly these actions if you run "terraform apply" now.
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $
```
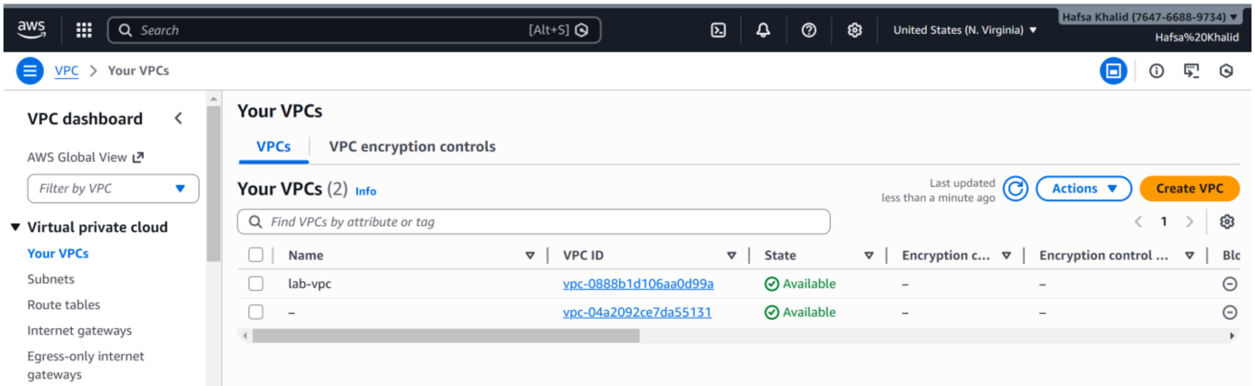
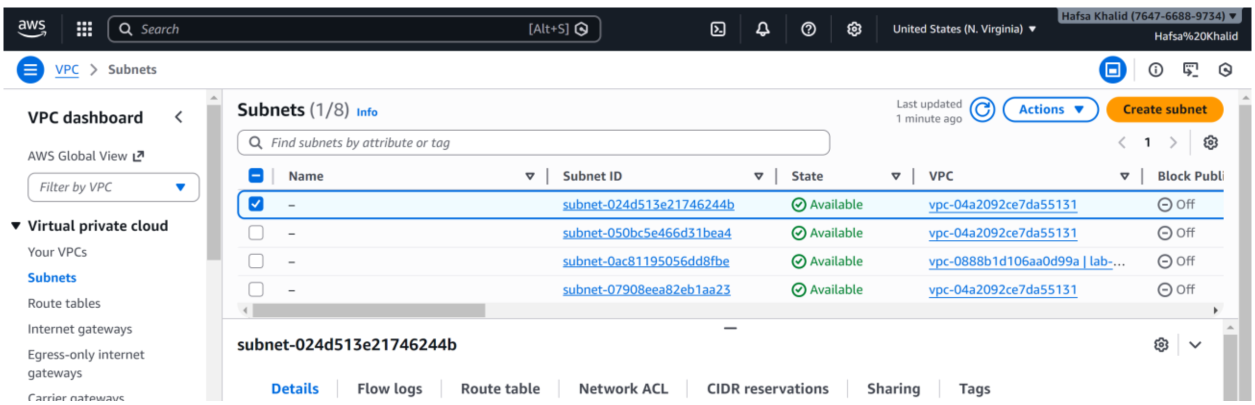
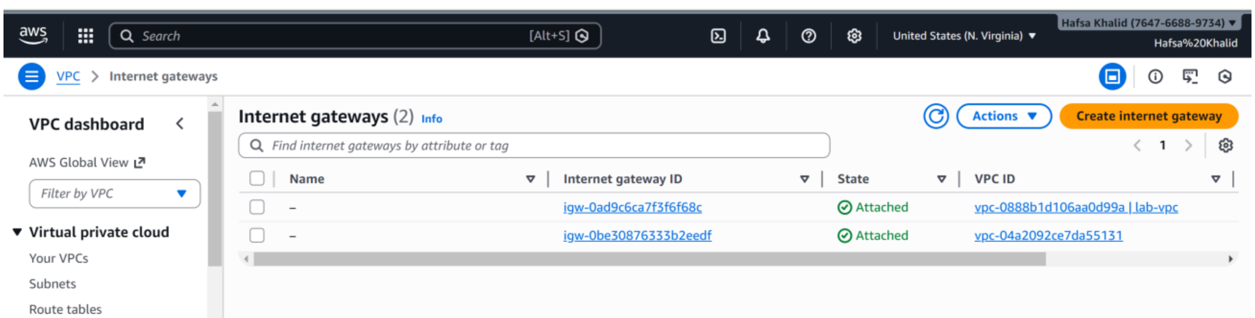
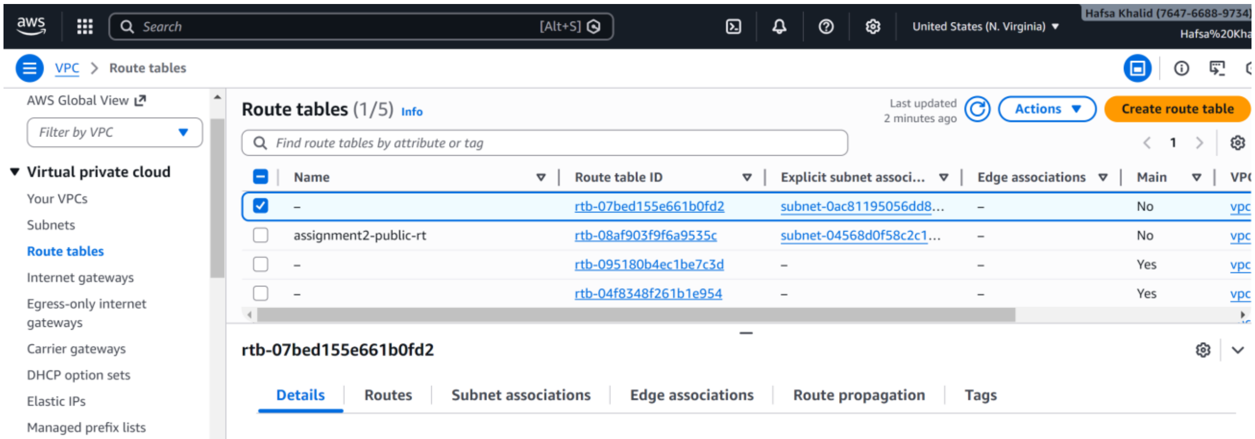

```
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $ terraform apply
    }

Plan: 1 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + ec2_public_ip = (known after apply)

Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes

aws_instance.myapp_ec2: Creating...
aws_instance.myapp_ec2: Still creating... [00m10s elapsed]
aws_instance.myapp_ec2: Creation complete after 14s [id=i-07b68bf175df7e9b2]

Apply complete! Resources: 1 added, 0 changed, 0 destroyed.

Outputs:

ec2_public_ip = "51.112.50.128"
@Hafsa-Khalid021 ➜ /workspaces/Lab-Exam (main) $
```

```
@Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $ terraform output
ec2_public_ip = "51.112.50.128"
@Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $
```









Q3 – Ansible Playbook for EC2 Web Server Using Q2 Instance

```
@Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $ ansible-playbook -i hosts nginx.yml --user
 ec2-user --private-key ~/.ssh/id_ed25519

PLAY [Install nginx on EC2] *************************************************************

TASK [Gathering Facts] ******************************************************************
[WARNING]: Host '51.112.50.128' is using the discovered Python interpreter at '/usr/bin/pyt
hon3.9', but future installation of another Python interpreter could cause a different inte
rpreter to be discovered. See https://docs.ansible.com/ansible-core/2.20/reference_appendic
es/interpreter_discovery.html for more information.
ok: [51.112.50.128]

TASK [Install nginx] ********************************************************************
ok: [51.112.50.128]

TASK [Start nginx] **********************************************************************
ok: [51.112.50.128]

PLAY RECAP ******************************************************************************
51.112.50.128              : ok=3    changed=0    unreachable=0    failed=0    skipped=0
 rescued=0    ignored=0

@Hafsa-Khalid021 →/workspaces/Lab-Exam (main) $
```


Not secure
51.112.50.128
This is Hafsa's Terraform environment